

Roll No.

Total Pages : 3

CMCS/M-22

24541

MACHINE LEARNING USING PYTHON

Paper–MS-20-42

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. (a) What do you mean by exception in a program?
- (b) Comment on the need of matplotlib.
- (c) What are the various parameters needed to implement Random Forest using Scikit Learn?
- (d) Define well-posed learning problems.
- (e) What is inductive bias in decision tree learning?
- (f) Differentiate between Lazy and Eager learning.

6×2½=15

UNIT-I

2. (a) Discuss the various operations that can be performed on dictionaries in python. Comment on the scope and lifetime of variables used in Python. 7½

- (b) Assume that a file contains integers separated by newlines. Write a code segment that opens the file and prints the average value of the integers. 7½

3. (a) What are map, reduce and filter functions in Python? Explain using suitable examples. 7½
- (b) Write a program in Python to remove all the duplicate words from a given text. 7½

UNIT-II

4. (a) What are the various data types in NumPy? Explain using suitable examples. 7½
- (b) What are the various steps followed in Pandas to clean and analyze the data? Explain in brief. 7½
5. What is Linear Regression? How can you visualize and implement linear regression using Scikit-learn? Explain in detail. 15

UNIT-III

6. (a) Write and explain the Find-S algorithm in detail. 7½
- (b) What are the various issues in decision tree learning? Explain in brief. 7½
7. How can you design a learning system? Explain in detail by designing a complete learning system of your own choice. 15

UNIT-IV

8. (a) Explain the concepts of Bayes optimal classifier in detail. $7\frac{1}{2}$
- (b) Write an explain working of EM algorithm. $7\frac{1}{2}$
9. (a) What is Vapnik-Chervonenkis dimension? Explain using suitable examples in any two applications. $7\frac{1}{2}$
- (b) Write and explain KNN algorithm in detail. $7\frac{1}{2}$